Sales Data Analysis and Business Insights

Data Science Final Project Report

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Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Streamlit

1. Problem Statement

Businesses record thousands of sales transactions, but raw records alone do not show which products, categories and regions drive revenue and profit, where discounting erodes margin, or how demand changes over time. Without a structured analysis, decisions on pricing, inventory and marketing rely on intuition. This project turns raw sales data into clean, reliable KPIs, visual analysis and an interactive dashboard so that decision-makers can find trends, top performers and problem areas quickly.

2. Introduction

The project is an end-to-end data science workflow: loading a real-world retail sales dataset, cleaning it, exploring it statistically and visually, calculating business KPIs, and presenting the findings in a Streamlit dashboard. The pipeline adapts to the columns available in the dataset, so optional analyses (time trends, customer segmentation) run only when the required fields exist.

Objectives

- Load and clean the sales dataset, handling missing values, duplicates and invalid records.
- Perform exploratory data analysis and compute summary statistics and KPIs.
- Analyse sales and profit by product, category, region and (where available) time period.
- Identify top-performing products and loss-making areas, and the effect of discounts.
- Present results through charts and an interactive, filterable dashboard.

3. Dataset and Methodology

Dataset: superstore.csv - 9,994 raw records, 13 columns (ship_mode, segment, country, city, state, postal_code, region, category, sub_category, sales, quantity, discount, profit).

Data cleaning summary

Step	Result
Raw records	9,994
Duplicate rows removed	17
Rows dropped (missing date/sales)	0
Outlier records flagged (IQR, kept)	1,167
Clean records used for analysis	9,977

Method

- **Cleaning:** standardised column names, removed duplicates, converted data types, dropped rows missing critical fields, filled other gaps (median for numbers, 'Unknown' for text), removed invalid values, flagged outliers with the IQR rule.
- **Feature engineering:** year, month, quarter, weekday and profit margin where the data allows.
- **KPIs:** total sales, profit, profit margin, orders, average order value, growth.

- **Analysis:** group summaries, discount impact, correlation, distributions, and RFM customer segmentation when customer IDs and dates exist.
- **Dashboard:** Streamlit app with filters, KPI cards, tabbed charts and CSV download.

Note: this dataset has no order-date column, so time-trend analysis is not included.

4. App UI Screenshots

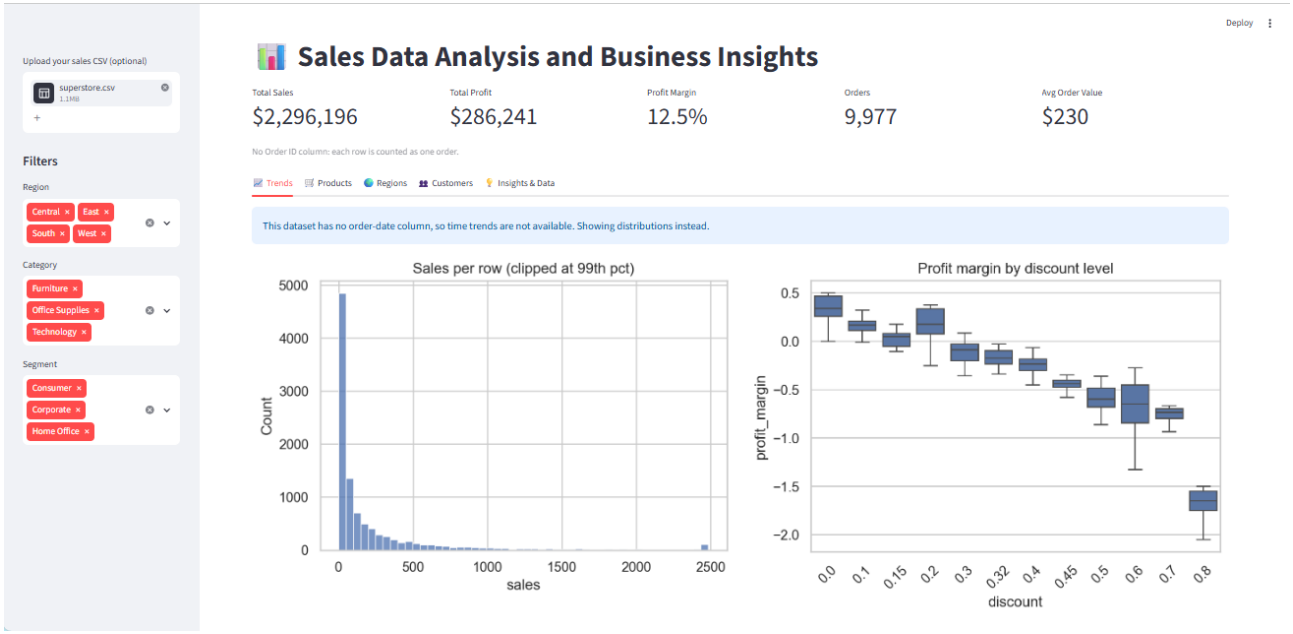


Figure: 1

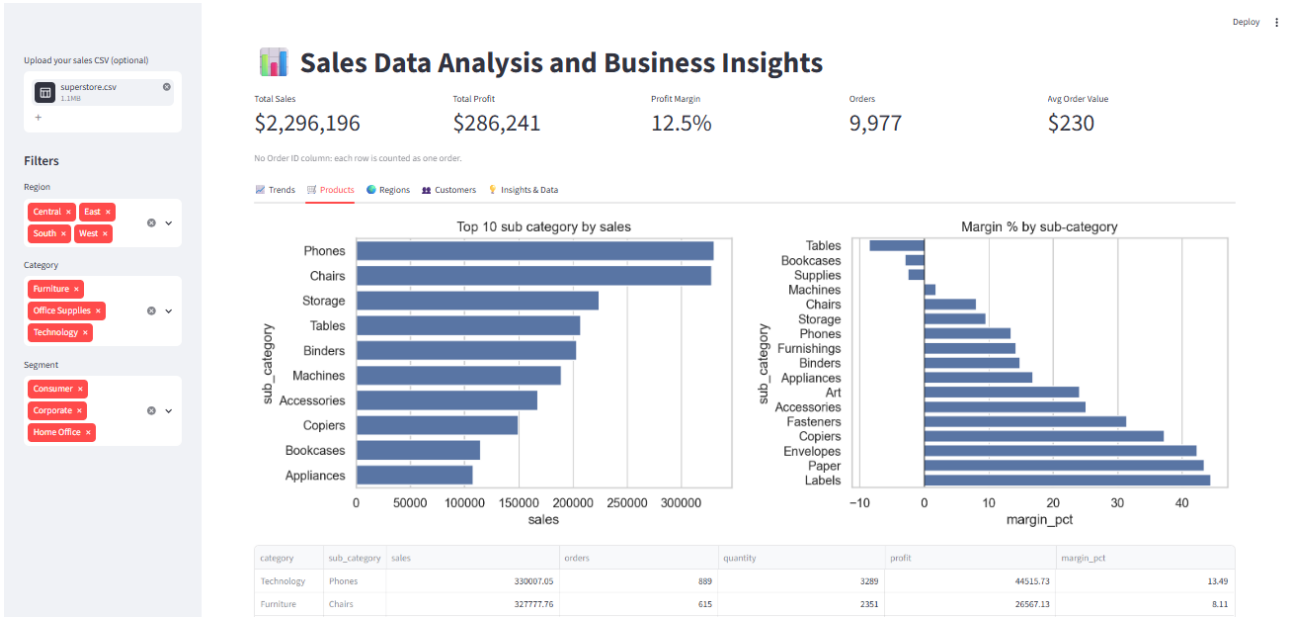
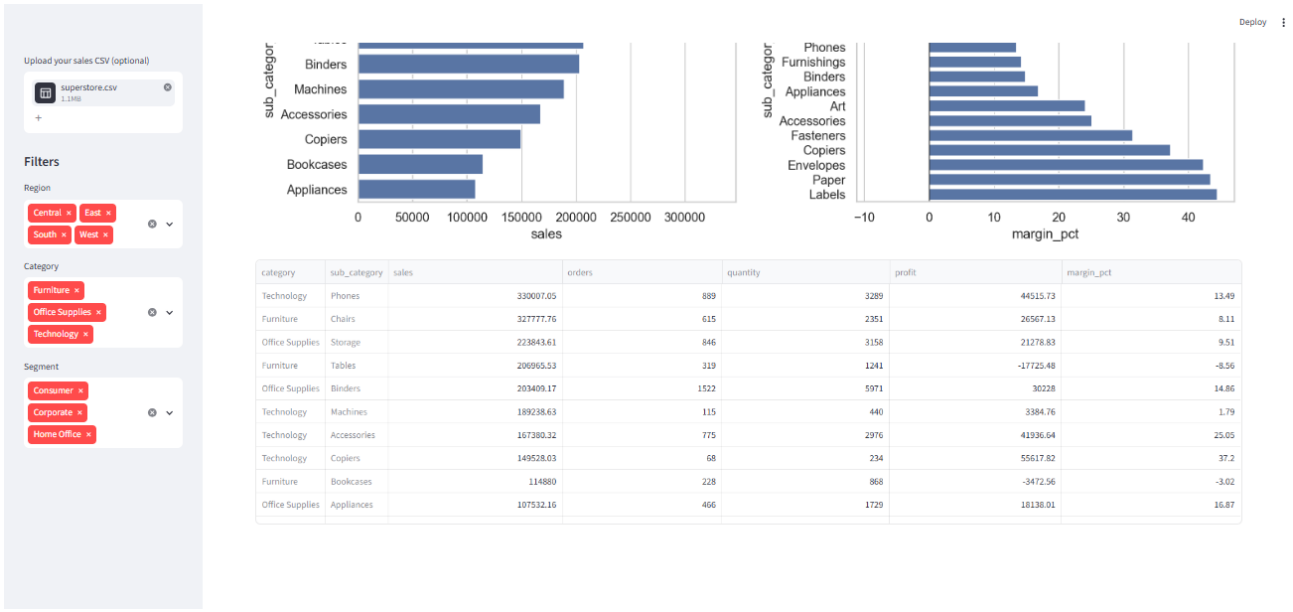


Figure: 2 (1)



category	sub_category	sales	orders	quantity	profit	margin_pct
Technology	Phones	330007.05	889	3289	44515.73	13.49
Furniture	Chairs	327777.76	615	2351	26567.13	8.11
Office Supplies	Storage	223843.61	846	3158	21278.83	9.51
Furniture	Tables	206965.53	319	1241	-17725.48	-8.56
Office Supplies	Binders	203409.17	1522	5971	30228	14.86
Technology	Machines	189238.63	115	440	3384.76	1.79
Technology	Accessories	167380.32	775	2976	41936.64	25.05
Technology	Copiers	149528.03	68	234	55617.82	37.2
Furniture	Bookcases	114880	228	868	-3472.56	-3.02
Office Supplies	Appliances	107532.16	466	1729	18138.01	16.87

Figure: 2 (2)

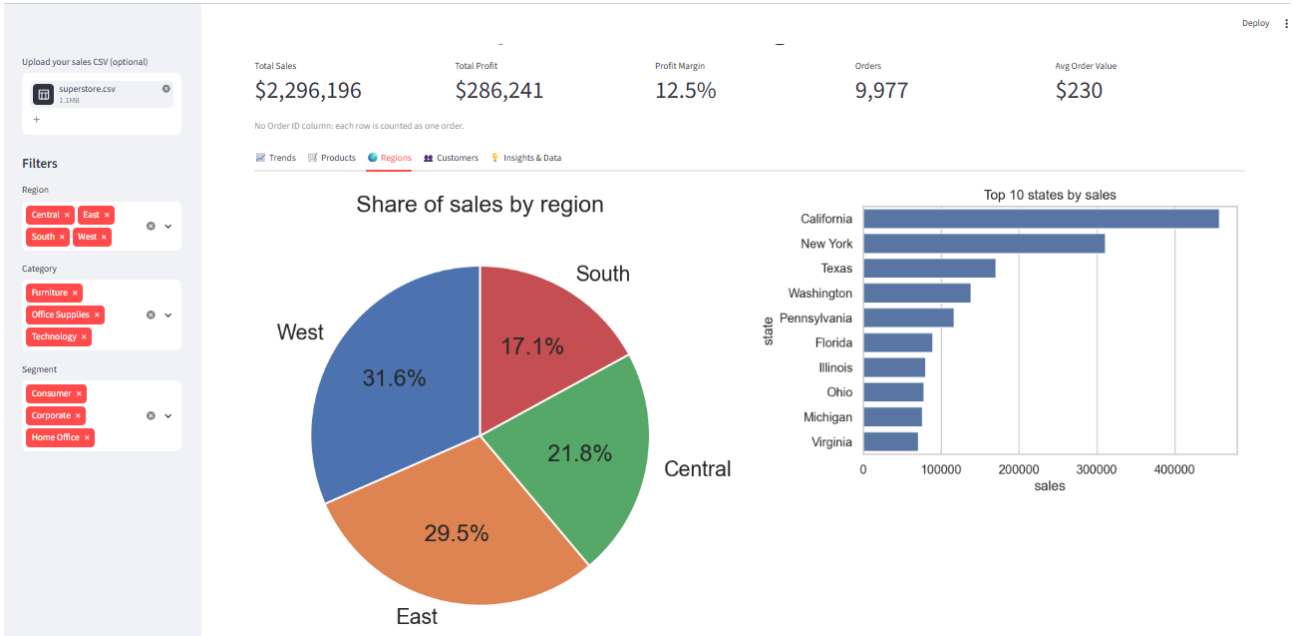


Figure: 3 (1)

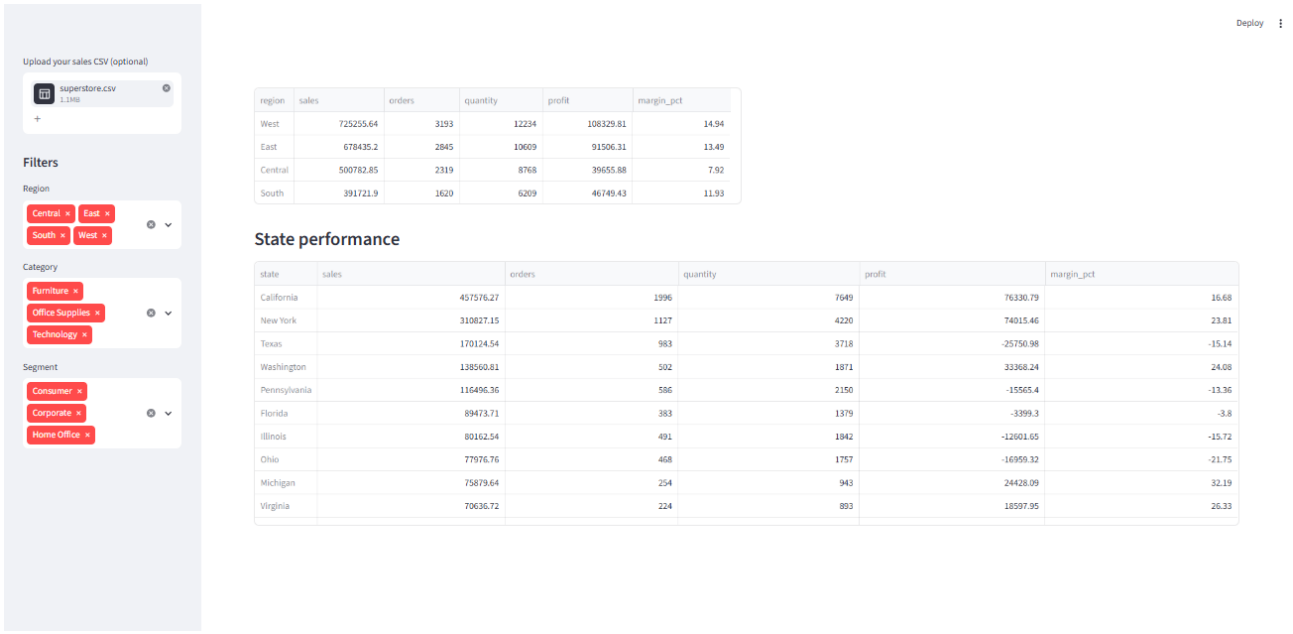


Figure: 3 (2)

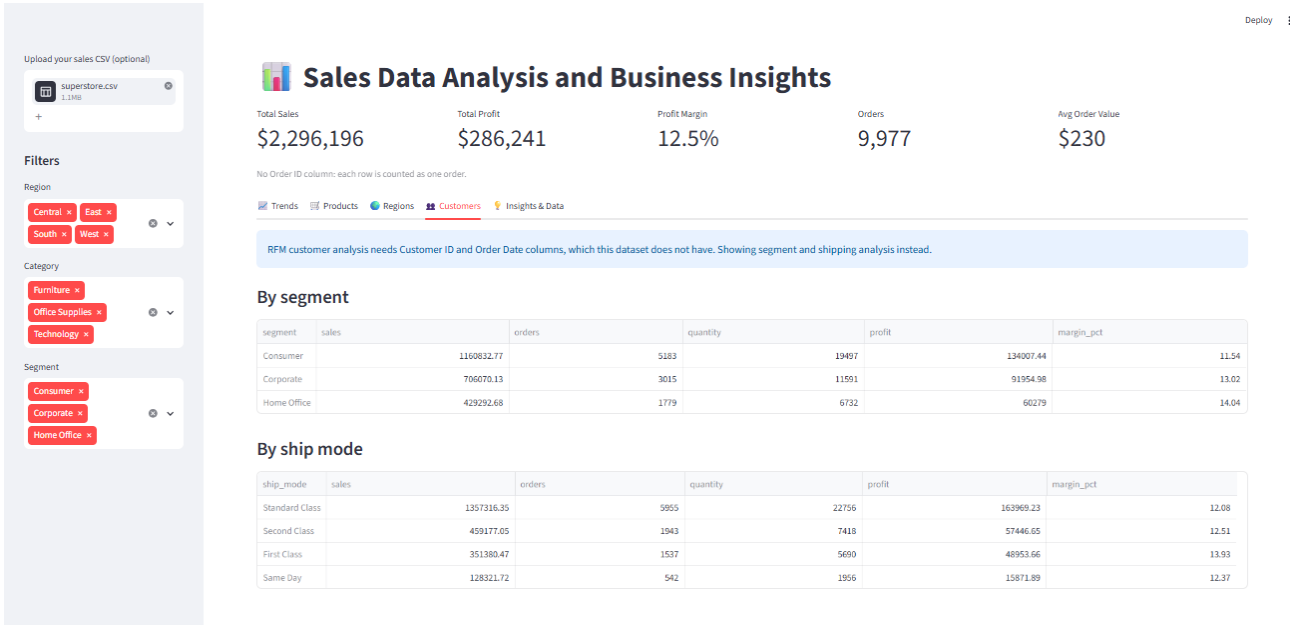


Figure: 4

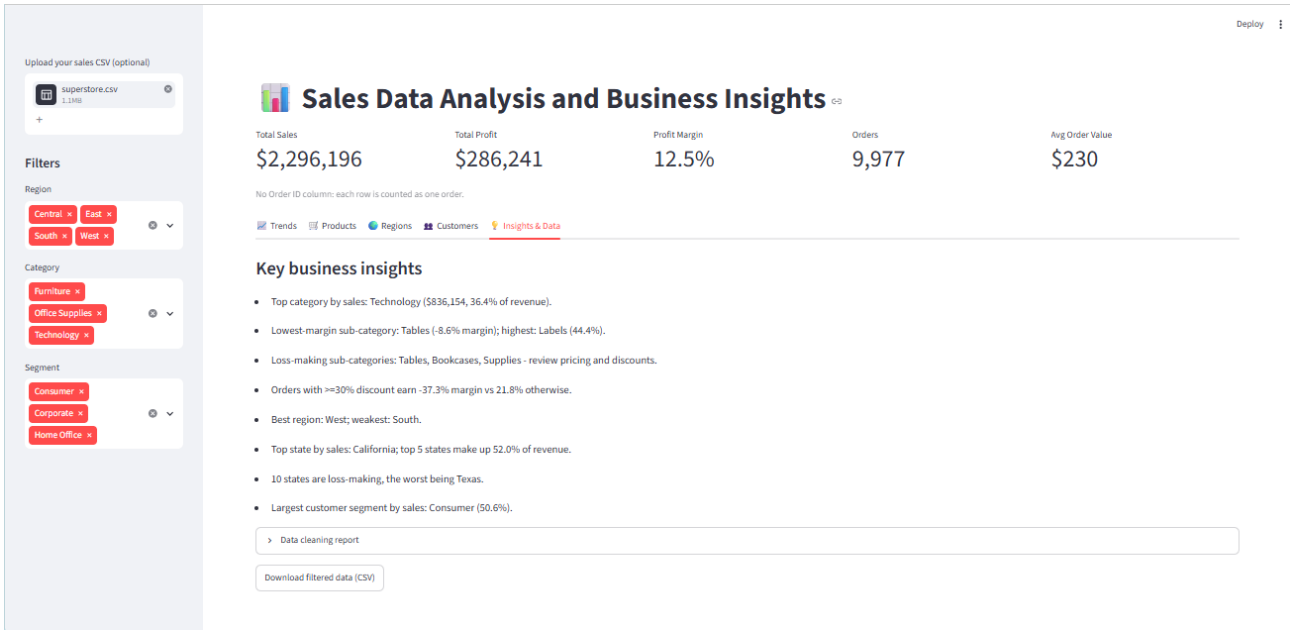

Deploy
Upload your sales CSV (optional)
superstore.csv
1.1MB
Filters
Region
Central
East
South
West
Category
Furniture
Office Supplies
Technology
Segment
Consumer
Corporate
Home Office
Sales Data Analysis and Business Insights
Total Sales
$2,296,196
Total Profit
$286,241
Profit Margin
12.5%
Orders
9,977
Avg Order Value
$230
No Order ID column: each row is counted as one order.
Trends
Products
Regions
Customers
Insights & Data
Key business insights
Top category by sales: Technology ($836,154, 36.4% of revenue).
Lowest-margin sub-category: Tables (-8.6% margin); highest: Labels (44.4%).
Loss-making sub-categories: Tables, Bookcases, Supplies - review pricing and discounts.
Orders with >=30% discount earn -37.3% margin vs 21.8% otherwise.
Best region: West; weakest: South.
Top state by sales: California; top 5 states make up 52.0% of revenue.
10 states are loss-making, the worst being Texas.
Largest customer segment by sales: Consumer (50.6%).
Data cleaning report
Download filtered data (CSV)

Figure: 5

5. Results

Key performance indicators

KPI	Value
Total sales	$2,296,196
Orders / records	9,977
Average order value	$230.15
Units sold	37,820
Total profit	$286,241
Profit margin	12.5%

Key business insights

- Top category by sales: Technology ($836,154, 36.4% of revenue).
- Lowest-margin sub-category: Tables (-8.6% margin); highest: Labels (44.4%).
- Loss-making sub-categories: Tables, Bookcases, Supplies - review pricing and discounts.
- Orders with >=30% discount earn -37.3% margin vs 21.8% otherwise.
- Best region: West; weakest: South.
- Top state by sales: California; top 5 states make up 52.0% of revenue.
- 10 states are loss-making, the worst being Texas.
- Largest customer segment by sales: Consumer (50.6%).

Performance by category

Category	Sales ($)	Profit ($)	Margin (%)	Records
Technology	836,154	145,455	17.4	1,847
Furniture	741,306	18,422	2.5	2,118
Office Supplies	718,735	122,365	17.0	6,012

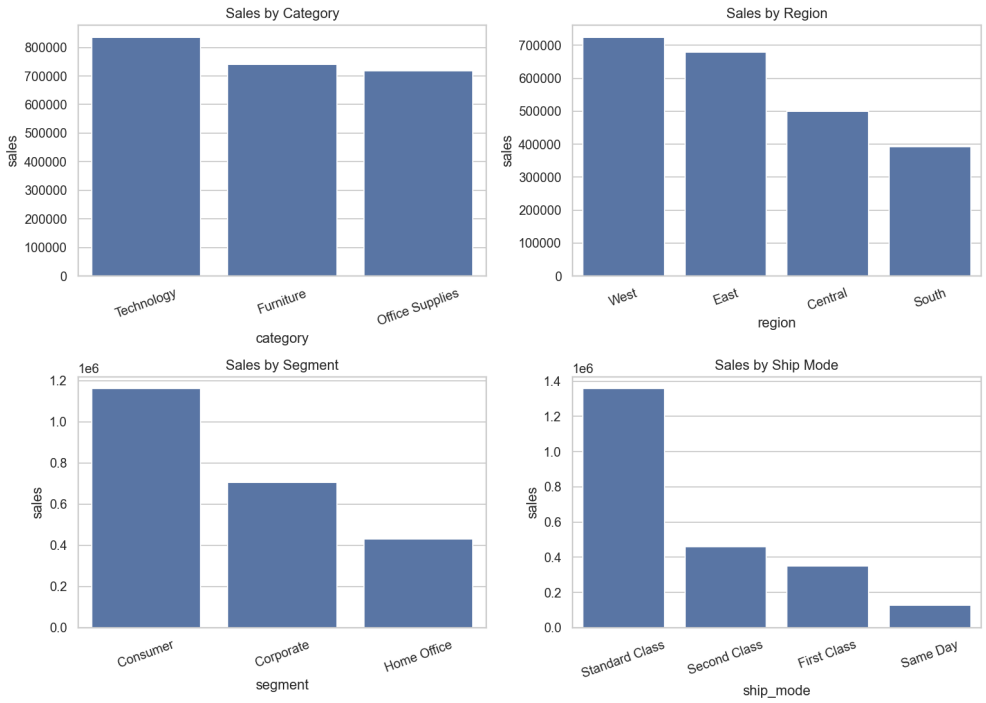
Performance by region

Region	Sales ($)	Profit ($)	Margin (%)	Records
West	725,256	108,330	14.9	3,193
East	678,435	91,506	13.5	2,845
Central	500,783	39,656	7.9	2,319
South	391,722	46,749	11.9	1,620

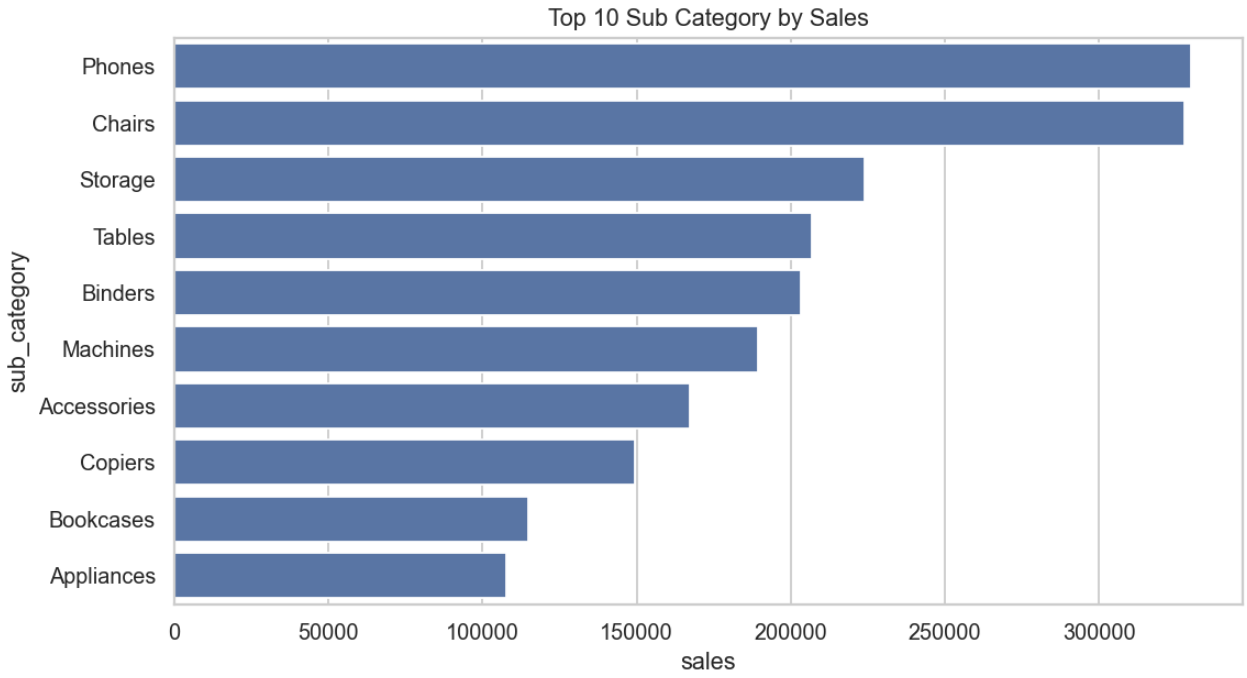
Performance by sub category

Sub Category	Sales ($)	Profit ($)	Margin (%)	Records
Phones	330,007	44,516	13.5	889
Chairs	327,778	26,567	8.1	615
Storage	223,844	21,279	9.5	846
Tables	206,966	-17,725	-8.6	319
Binders	203,409	30,228	14.9	1,522
Machines	189,239	3,385	1.8	115
Accessories	167,380	41,937	25.1	775
Copiers	149,528	55,618	37.2	68
Bookcases	114,880	-3,473	-3.0	228
Appliances	107,532	18,138	16.9	466

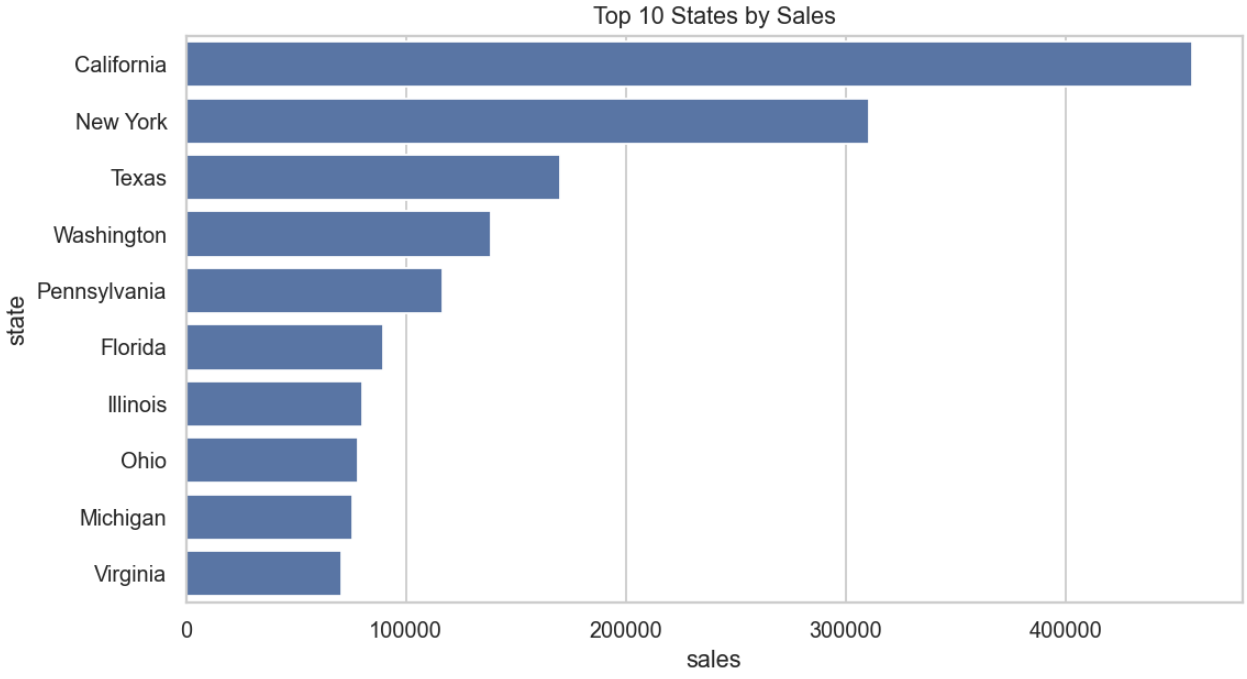
Visual analysis



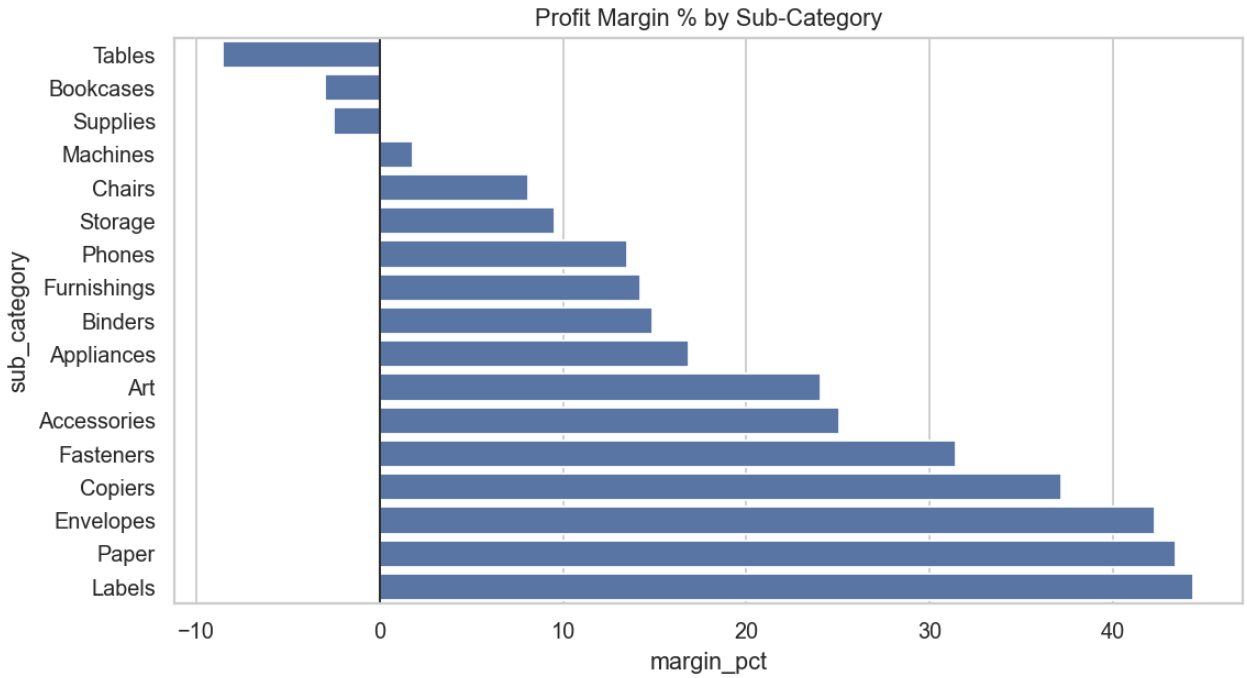
Sales by category, region, segment and shipping mode.



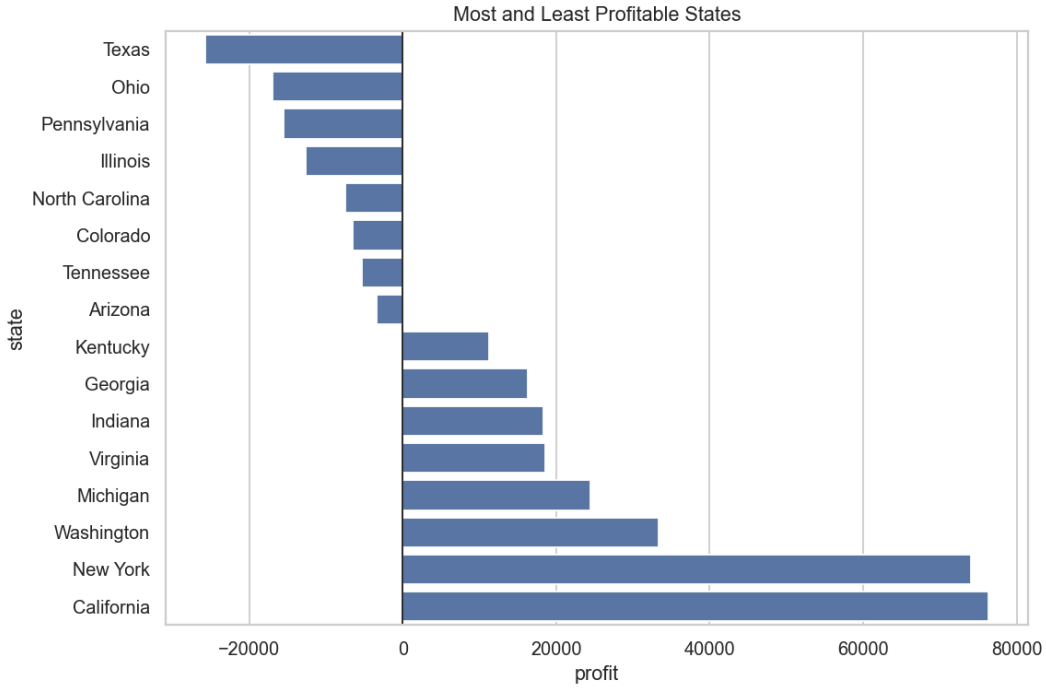
Top 10 products / sub-categories by sales.



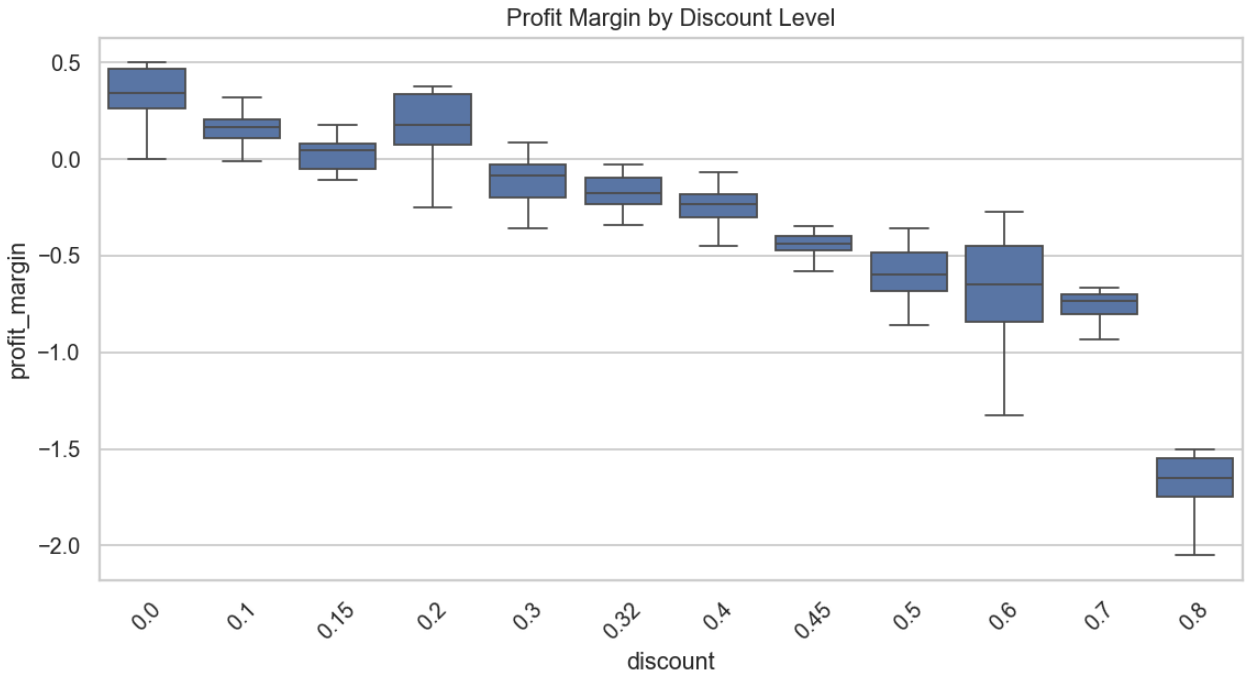
Top 10 states by sales.



Profit margin (%) by sub-category.



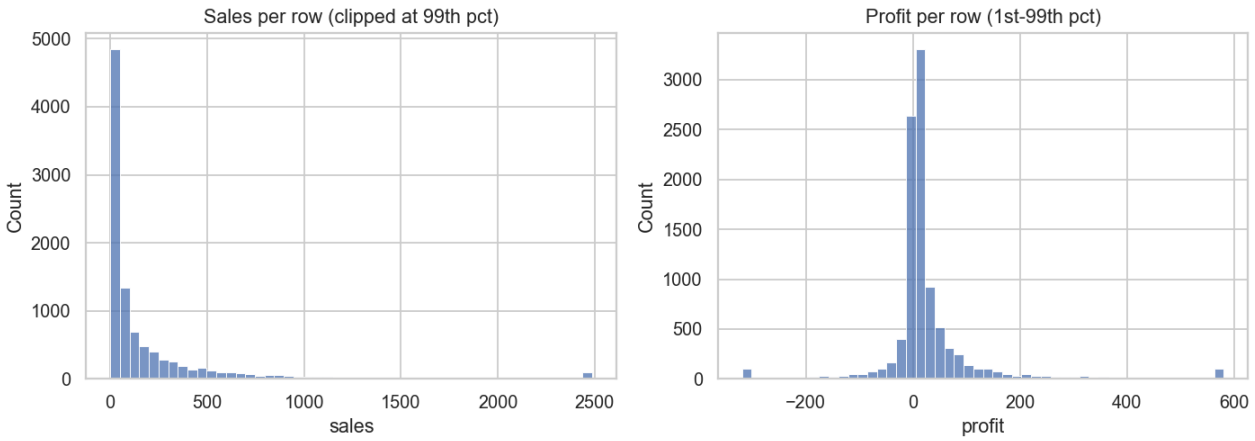
Most and least profitable states.



Profit margin by discount level.



Correlation between numeric variables.



Distribution of sales and profit per record.

6. Conclusion

The project shows how a systematic cleaning and analysis pipeline converts raw sales records into decisions: it highlights the strongest categories and regions, exposes loss-making segments, and quantifies how heavy discounting affects profit margin. The dashboard lets non-technical users filter and explore these results on their own.

7. Future Approach

- **Sales forecasting:** add time-series models (ARIMA / Prophet) to predict future demand when date data is available.
- **Profit prediction and discount optimisation:** train regression or gradient-boosting models to find discount levels that protect margin.
- **Customer analytics:** extend RFM into customer lifetime value, churn prediction and product recommendations using customer-level data.
- **Richer interactivity:** move charts to Plotly for hover, zoom and drill-down.
- **Live data pipeline:** connect the dashboard to a database or API instead of a static CSV, with scheduled refresh.
- **Quality and operations:** add automated tests, CI checks, authentication and automated report emails.